STATISTICS FOR DATA ANALYTICS & DATA SCIENCE

MODULE 1: STATISTICS FUNDAMENTALS

1. What is Statistics?

Definition:
Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data to make better decisions.

In Simple Words:
Statistics helps us understand data and extract meaningful insights so we can make informed decisions.

Why Do We Need Statistics?

- Data is everywhere in business and life.
- Raw data is useless unless we understand it.
- Statistics transforms data into information, insights, and knowledge.
- It helps reduce uncertainty and supports decision making.

Real Life Example

A company collects sales data from different regions. Statistics helps them understand where sales are high/low, why sales change, and what actions to take to improve.

Where to Apply?

- Business Analytics
- Finance
- Operations
- HR Analytics
- Data Science

2. Importance of Statistics in Data Analytics

- Converts raw data into meaningful insights.
- Helps in identifying trends, patterns, and relationships.
- Supports data-driven decision making.
- Helps in forecasting future outcomes.
- Reduces risk and uncertainty.
- Essential for building models in Data Science.
- Important for A/B testing, surveys, experiments, and research.

Example

An e-commerce company wants to know if a new website design increases sales. Statistics helps them test the result and decide whether to implement the change for all users.

3. Applications in Real Life Business

Sales & Marketing

Analyze sales trends, customer behavior, campaign performance.

Finance

Risk analysis, investment return, fraud detection.

Operations

Process improvement, quality control, demand forecasting.

HR Analytics

Employee performance, attrition analysis, salary benchmarking.

Data Science

Model building, predictions, machine learning, AI.

4. Population vs Sample

Population:
The entire group of individuals or items that we are interested in studying.

Example: All customers of Amazon in India.

Sample:
A subset or a part of the population selected for study.

Example: 1,000 customers selected from Amazon India customers.

Key Point:
Studying the entire population is often costly, time-consuming, or impossible. So we study a sample and make conclusions about the population.

5. Parameter vs Statistic

Parameter (Population Measure)	Statistic (Sample Measure)
Calculated from the entire population.	Calculated from a sample.
Fixed value (does not change).	Can change from sample to sample.
Represented by Greek letters (μ, σ, $\bar{P}$).	Represented by English letters ($\bar{x}$, s, $\hat{\bar{p}}$).
Example: Population mean (μ), Population standard deviation (σ).	**Example:** Sample mean ($\bar{x}$), Sample standard deviation (s).

Real Life Example

The average salary (μ) of all employees in a company is a parameter.

The average salary ($\bar{x}$) of 500 employees taken randomly is a statistic.

6. Census vs Sampling

Census:
Collecting data from every member of the population.

Example: Counting every vehicle in a city.

Sampling:
Collecting data from a sample and estimating the population.

Example: Surveying 1,000 people to estimate city population.

When to Use?

- ✓ **Use Census** → When population is small, or high accuracy is required.
- ✓ **Use Sampling** → When population is large, or resources are limited.

12. Discrete vs Continuous Data

Discrete Data	Continuous Data
• Countable values. • Usually whole numbers. • There is a gap between two values. **Example:** Number of Students in a class (35, 36, 37...) Number of Cars in a parking lot (0, 1, 2, 3...)	• Can take any value within a range. • Measured values. • No gap between two values. **Example:** Height of a person (150.25 cm, 150.26 cm...) Weight of a person (65.32 kg, 65.33 kg...)

Real Life Example

- Discrete: Number of orders placed on an e-commerce website in a day.
- Continuous: Amount spent by a customer on the website.

Why It Matters?

The type of data decides:

- ✓ Which statistical methods to use
- ✓ How to summarize the data
- ✓ How to visualize the data

13. Levels of Measurement (Scales of Measurement)

Level	Description	Example	Properties	Analysis You Can Do
Nominal	Data used to name or categorize. No order.	Gender (Male, Female), Blood Group (A, B, O, AB)	Categories only No order	Count, Percentage, Mode, Charts
Ordinal	Data has order or ranking, but difference between values is not equal.	Education Level (High School, Graduate, Postgraduate), Customer Satisfaction (Poor, Average, Good, Excellent)	Order or Rank No equal intervals	Median, Percentiles, Rank, Charts
Interval	Data has equal intervals but no true zero.	Temperature in °C or °F, Calendar Years	Equal intervals No true zero	Mean, Standard Deviation, Correlation, Regression
Ratio	Data has equal intervals and a true zero.	Height, Weight, Age, Income, Sales, Distance	Equal intervals True zero exists	All statistical analysis (Mean, SD, CV, etc.)

Real Life Example

- Nominal: Types of products (Mobile, Laptop, Tablet)
- Ordinal: Product ratings (1 Star, 2 Star, 3 Star, ...)
- Interval: Temperature in °C
- Ratio: Salary in ₹, Sales Amount in ₹

Key Takeaway

Higher the level of measurement, more powerful the statistical analysis.

14. Primary vs Secondary Data

Primary Data	Secondary Data
• Collected first-hand by the researcher. • Original and specific to the objective. • More accurate and reliable. • Time-consuming and costly. **Examples:** Surveys, Interviews, Observations, Experiments, Questionnaires	• Collected by someone else. • Already available. • Less costly and time-saving. • May not be 100% accurate for your need. **Examples:** Government reports, Company records, Research papers, News articles, Websites

Real Life Example

A company wants market size data:

- Primary: Conducting a survey of 2,000 customers.
- Secondary: Using data from government reports or market research reports.

When to Use?

- ✓ Use Primary Data → When accuracy and specific information is critical.
- ✓ Use Secondary Data → When quick insights are needed or budget is limited.

15. Data Collection Methods

Surveys
Collect data using structured questionnaires.
Example: Customer satisfaction survey.

Interviews
Collect data by asking questions directly.
Example: HR interviews.

Observation
Collect data by observing behavior or events.
Example: Observing customer behavior in a store.

Experiment
Collect data by testing under controlled conditions.
Example: A/B testing of a website.

Case Study
In-depth study of an individual, group or organization.
Example: Study of a successful startup.

Existing Records
Collect data from files, databases or documents.
Example: Sales records, CRM data.

Key Point: The quality of analysis depends on the quality of data collected.
Always ensure data is relevant, accurate, complete, consistent, and collected ethically.

16. Sampling Basics & Sampling Techniques

What is Sampling?

Sampling is the process of selecting a part (sample) of the population to represent the whole population.



Why Do We Use Sampling?

- ✓ Saves time, cost, and resources.
- ✓ Feasible for large or infinite populations.
- ✓ Less data to collect and analyze.
- ✓ Can provide accurate results if done correctly.

Common Sampling Techniques

1. Simple Random Sampling
Every member has an equal chance of being selected.

2. Systematic Sampling
Selecting every k^{th} item after a random starting point.

3. Stratified Sampling
Population is divided into subgroups (strata) and samples are taken from each group.

4. Cluster Sampling
Population is divided into clusters, random clusters are selected, and all members in those clusters are studied.

5. Convenience Sampling
Samples are selected based on ease of access.

Real Life Example

- A university wants to know the average monthly expense of students.

→ **Population:** All 20,000 students in the university.

→ **Sample:** 400 students selected using stratified sampling (by department).

→ **Result:** The sample data is used to estimate the average monthly expense of all students.

Key Takeaway

Good sampling leads to good results. Poor sampling leads to wrong conclusions.

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17. Sampling Bias & Sampling Error

Sampling Bias

Sampling bias occurs when the sample selected is not representative of the population.

Types of Sampling Bias:

- **Selection Bias:** Some groups have higher chance of being selected.
- **Non-response Bias:** Some selected people do not respond.
- **Coverage Bias:** Some parts of the population are left out.
- **Voluntary Response Bias:** People with strong opinions are more likely to respond.



Sampling Error

Sampling error is the difference between the sample result and the true population value.

Causes of Sampling Error:

- Random chance
- Sample size
- Variation in the population
- Measurement errors



Real Life Example

A poll predicts 60% people will vote for a candidate. Actual election result is 55%.

Difference = 5%

This difference is called Sampling Error.

How to Reduce Bias & Error?

- ✓ Use random sampling techniques.
- ✓ Increase sample size.
- ✓ Ensure sample represents all important groups.
- ✓ Use proper data collection methods.
- ✓ Check and clean data before analysis.

Example

A company surveys 300 customers from different cities, age groups, and income levels randomly.

→ **Bias Reduced** because the sample represents all types of customers.

→ **Error Reduced** because the sample size is adequate and random.

Key Takeaway

Bias affects the accuracy of data.
Error affects the precision of data.

Both should be minimized for better statistical analysis.

MODULE 1: STATISTICS FUNDAMENTALS — COMPLETE!

You Learned:
The basic building blocks of Statistics.

You Understood:
Data, Types, Collection, Population, Sampling and more.

You Can Now:
Identify data types, choose sampling techniques and avoid common mistakes.

Why It Matters:
These fundamentals are the base for all advanced Statistical methods, Data Analytics and Data Science.

Next Steps:
In the next modules we will apply these concepts to perform Descriptive Statistics and beyond.

Module 1 Summary

- Understand what statistics is and why it is important.
- Know the difference between population and sample.
- Understand parameters and statistics.
- Learn data types and measurement scales.
- Understand sampling and its importance.
- Know common biases and errors in sampling.

> "Statistics gives us the power to turn data into decisions.
> Better Data. Better Insights. Better Decisions."

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STATISTICS FOR DATA ANALYTICS & DATA SCIENCE

MODULE 1: STATISTICS FUNDAMENTALS

18. Summary: Key Differences at a Glance

Concept	Term 1	vs	Term 2	Key Difference
Population	Entire group of interest	vs	Sample	Population is whole, Sample is a part.
Parameter	Population measure	vs	Statistic $\bar{x}$	Parameter describes population, Statistic describes sample.
Census	Data from every member	vs	Sampling	Census is complete, Sampling is partial.
Qualitative Data	Categories, labels	vs	Quantitative Data	Qualitative is non-numeric, Quantitative is numeric.
Discrete Data	Countable, whole numbers	vs	Continuous Data	Discrete has gaps, Continuous has no gaps.
Levels of Measurement	Nominal, Ordinal, Interval, Ratio	vs	–	Higher levels provide more information.
Primary Data	Collected first-hand	vs	Secondary Data	Primary is original, Secondary is already available.
Data Collection Methods	Surveys, Interviews, Observation, etc.	vs	–	Different methods suit different situations.
Sampling Bias	Bias in selecting sample	vs	Sampling Error	Bias is systematic error, Error is random difference.

Real Life Integrated Example

A retall company wants to know the average spending of customers per month.

- **Population:** All customers of the company (5 lakh customers)
- **Sample:** 2,000 customers selected randomly
- **Parameter:** True average spending of all 5 lakh customers (μ)
- **Statistic:** Average spending of 2,000 customers in the sample ($\bar{x}$)
- **Data Type:** Quantitative Continuous (spending can be any value)
- **Data Collection Method:** Online survey & transaction records
- **Bias:** If only high-value customers are selected (Selection Bias)
- **Error:** Due to sample size or random variation (Sampling Error)

Key Insight

Good statistics starts with correct data, proper sampling, the right method, and clear understanding.

19. Module 1 Summary – Statistics Fundamentals

1 What is Statistics?
It is the science of collecting, organizing, analyzing, interpreting, and presenting data.

2 Importance of Statistics
Helps in understanding data, identifying patterns, reducing uncertainty, and making data-driven decisions.

3 Applications in Business
Used in Sales, Marketing, Finance, Operations, HR Analytics, and Data Science.

4 Population vs Sample
Population is the entire group, sample is a part selected for study.

5 Parameter vs Statistic
Parameter describes the population, statistic describes the sample.

6 Census vs Sampling
Census collects data from all, sampling collects data from a part.

7 Variables & Observations
Variables are characteristics, observations are the recorded values.

8 Types of Data
Qualitative (categories) and Quantitative (numbers). Quantitative can be Discrete or Continuous.

9 Levels of Measurement
Nominal → Ordinal → Interval → Ratio (increasing level of information and analysis power).

10 Primary vs Secondary Data
Primary data is original, specific, and reliable. Secondary data is available, less costly, but may not be 100% accurate.

11 Data Collection Methods
Surveys, Interviews, Observation, Experiments, Case Study, and Existing Records.

12 Sampling Basics
Sampling helps study large populations efficiently with less time and cost.

13 Sampling Techniques
Simple Random, Systematic, Stratified, Cluster, and Convenience Sampling.

14 Sampling Bias
Occurs when the sample is not representative of the population.

15 Sampling Error
Occurs due to chance. Can be reduced by increasing sample size and using random techniques.

16 Why It Matters?
Correct understanding of fundamentals ensures accurate analysis, reliable insights, and better decision making.

FINAL TAKEAWAY

Understanding fundamentals builds the foundation for all advanced statistical techniques and models.

- **Right Data** – Collect the right data.
- **Right Method** – Use the right statistical method.
- **Right Decision** – Make the right business decision.

> "No matter how advanced the analysis is, everything starts with strong fundamentals."